

PALMYRA, Syrian Arab Republic

WMO: 400610

Lat: 34.557N

Lon: 38.317E

Elev: 403

StdP: 96.58

Time Zone: 2.00 (SYR)

Period: 94-15

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-0.3	1.0	-11.2	1.5	8.7	-9.1	1.8	8.9	8.0	9.0	6.3	8.6	0.6	270	0.399

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.7	41.0	20.3	39.5	19.8	38.2	19.5	22.7	34.1	21.8	33.2	21.1	32.4	1.4	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.5	15.0	26.6	18.7	14.2	26.2	17.9	13.5	25.8	69.4	33.7	65.9	33.0	63.4	32.4	30.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.1	6.8	5.7	-2.6	43.7	2.2	2.2	-4.2	45.3	-5.5	46.5	-6.8	47.8	-8.4	49.3		
(5)			-4.4	25.1	1.8	1.9	-5.7	26.5	-6.7	27.6	-7.7	28.7	-9.0	30.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	19.8	7.6	10.1	14.0	18.9	24.2	28.4	30.8	30.9	27.4	22.2	14.0	8.9		
	DBStd	8.83	2.58	2.90	3.63	3.67	3.32	2.52	2.08	2.02	2.28	3.14	3.28	2.85		
	HDD10.0	182	82	31	7	1	0	0	0	0	0	0	7	54		
	HDD18.3	1174	334	231	142	34	1	0	0	0	0	6	132	293		
	CDD10.0	3773	6	34	131	268	442	551	646	648	522	379	128	19		
	CDD18.3	1723	0	0	8	52	184	301	387	389	272	127	2	0		
	CDH23.3	21356	0	0	74	501	2056	3949	5424	5359	2998	980	14	0		
	CDH26.7	12230	0	0	18	163	996	2330	3441	3362	1590	329	1	0		
(14) Wind	WSAvg	1.8	1.0	1.3	1.7	2.1	2.2	2.6	3.0	2.7	2.0	1.2	0.9	0.9		
(15) Precipitation	PrecAvg	130	20	21	24	16	9	0	0	0	0	9	16	20		
	PrecMax	270	67	134	108	85	65	16	1	0	2	43	64	85		
	PrecMin	38	0	0	0	0	0	0	0	0	0	0	0	2		
	PrecStd	56	16	22	23	21	13	2	0	0	1	11	17	16		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.4	21.3	29.1	34.6	37.7	40.9	43.5	42.7	38.7	34.2	25.0	18.3		
		MCWB	10.5	11.5	13.7	16.5	18.8	19.4	20.0	21.1	20.2	18.7	14.4	11.9		
	2%	DB	14.8	18.6	25.2	30.9	35.4	38.9	41.3	40.9	36.8	32.0	22.9	16.1		
		MCWB	9.5	10.2	12.3	15.2	17.5	19.2	19.8	20.9	19.3	17.3	13.4	10.6		
	5%	DB	13.6	17.1	22.7	28.6	33.9	37.5	39.7	39.4	35.3	30.2	21.2	14.7		
		MCWB	8.9	9.8	11.6	14.6	16.9	18.6	19.6	20.0	18.8	16.8	13.0	9.9		
	10%	DB	12.1	15.5	20.5	26.3	32.1	36.1	38.2	38.0	33.9	28.4	19.6	13.4		
		MCWB	8.1	9.1	10.9	13.7	16.5	18.1	19.5	19.8	18.4	16.2	12.3	9.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.5	12.3	15.3	19.2	21.7	22.2	24.0	24.0	22.7	21.0	16.9	12.8		
		MCDB	15.1	18.5	22.8	29.1	32.3	36.5	36.6	36.2	33.6	29.9	21.5	15.4		
	2%	WB	10.3	11.1	14.0	16.7	19.3	20.7	22.3	22.8	21.4	19.1	15.5	11.7		
		MCDB	13.6	17.0	21.5	26.8	30.7	34.0	34.6	33.8	31.6	27.1	20.1	14.5		
	5%	WB	9.5	10.4	12.8	15.4	18.2	19.7	21.4	22.0	20.5	18.1	14.4	10.9		
		MCDB	12.5	15.8	20.7	26.1	29.9	32.6	33.7	33.0	29.8	26.5	18.9	13.5		
	10%	WB	8.7	9.7	11.8	14.5	17.3	19.0	20.8	21.4	19.8	17.4	13.4	10.0		
		MCDB	11.4	14.5	19.2	24.9	29.4	32.8	32.8	32.2	28.8	25.7	18.0	12.7		
(35) Mean Daily Temperature Range	MDBR		8.2	9.5	10.5	11.9	13.3	14.2	14.7	14.4	12.9	11.4	10.6	8.4		
	5% DB	MCDBR	9.9	11.5	13.4	14.4	15.0	15.5	16.0	15.3	14.5	13.4	12.4	10.3		
		MCWBR	5.6	5.5	5.3	5.5	5.1	4.3	3.6	3.3	4.3	5.1	6.2	5.9		
	5% WB	MCDBR	8.6	10.1	11.2	12.8	13.6	14.9	15.0	14.8	13.2	11.6	9.4	8.3		
		MCWBR	5.3	5.1	5.1	5.6	5.3	4.4	3.9	3.7	4.2	4.6	5.2	5.1		
(40) Clear-Sky Solar Irradiance	taub		0.363	0.396	0.435	0.499	0.496	0.432	0.426	0.432	0.422	0.454	0.390	0.370		
	taud		2.145	2.044	1.936	1.789	1.826	2.015	2.047	2.027	2.041	1.943	2.087	2.122		
	Ebn at Noon		816	830	830	795	803	854	856	844	832	756	774	778		
	Edh at Noon		119	147	178	217	211	175	168	168	158	161	125	115		
(44) All-Sky Solar Radiation	RadAvg		2.83	3.73	5.09	6.18	7.20	8.20	7.88	7.04	5.95	4.43	3.28	2.58		
	RadStd		0.14	0.28	0.21	0.27	0.35	0.19	0.16	0.23	0.23	0.23	0.23	0.19		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB		1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page